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Nationwide potential causal long-term effects of $PM_{2.5}$ component mixture on ischemic heart disease mortality in the United States, 2000–2019

Andong Huang [a,1], Zhiqiang Li [b,c,1], Xiaojing Song [d,1], Shao lin [e], Jiaqi Wei [a], Yanji Qu [f], Qiao Hu [g], Guanghui Dong [c], Man Zhang [h], Guanjun Wang [b,*], Wangjian Zhang [c,*], Xinxin Chen [a,*]

[a] Breast disease center, Second Affiliated Hospital of Guangzhou Medical University, Guangzhou 510261, China
[b] Qingdao Hospital, University of Health and Rehabilitation Sciences (Qingdao Municipal Hospital), Qingdao, Shandong 266000, China
[c] Department of Medical Statistics, School of Public Health, Sun Yat-sen University, Guangzhou, Guangdong 510080, China
[d] Qilu Hospital (Qingdao), Cheeloo College of Medicine, Shandong University, Qingdao, Shandong 266000, China
[e] Department of Environmental Health Sciences, School of Public Health, University at Albany, the State University of New York, Rensselaer, NY 12144, USA
[f] Department of Cardiovascular Epidemiology, Guangdong Cardiovascular Institute, Guangdong Provincial People's Hospital & Guangdong Academy of Medical Sciences, Guangzhou 510000, China
[g] Department of Ultrasound, People's Hospital of Guangxi Zhuang Autonomous Region, 6# Taoyuan Road, Nanning, China
[h] Department of nosocomial infection management, Beijing Ditan Hospital, Capital Medical University, Beijing 100015, China

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ABSTRACT

Background: Ischemic heart disease (IHD) is a leading cause of global mortality. Although fine particulate matter ($PM_{2.5}$) is a risk factor, the nationwide causal contributions of its chemical components and their mixture to IHD mortality remain unclear.

Methods: We conducted a nationwide study covering 3108 counties across the contiguous United States during 2000–2019. Using a difference-in-differences design with two-way fixed models, we estimated long-term potential causal effects of individual $PM_{2.5}$ components on IHD mortality and applied quantile g-computation to quantify the joint mixture effects and the weights of each component. We further evaluated the effect modification by demographic characteristics, socioeconomic factors, and seasonal temperatures.

Results: An interquartile-range increase in 5 $PM_{2.5}$ components was associated with 1.61–8.83% increases in IHD mortality. The component mixture was associated with a 0.92% (95% CI: 0.60, 1.25) increase in IHD mortality, with sulfate (weight: 43.69%) and nitrate (34.97%) contributing the largest positive weights. The association was greater among older adults (effect increase: 4.81%), Hispanic or Latino (3.83%), and in counties with higher poverty (0.30%), higher physical inactivity (0.92%), and warmer climates (0.33–0.51%). Conversely, Black or African American (effect decrease: 2.46%) and higher health insurance coverage (1.05%) appeared to attenuate this association.

Conclusions: This study provides potential causal links for the joint effects of $PM_{2.5}$ component mixtures on IHD mortality, with stronger associations among vulnerable subpopulations, and identifies sulfate and nitrate as critical components. These findings underscore the necessity of emission controls targeting toxic constituents alongside public health strategies designed to protect susceptible populations.

1. Introduction

Ischemic heart disease (IHD), which encompasses conditions such as myocardial infarction and chronic coronary syndromes, is the leading cause of death worldwide and imposes a substantial public health burden (Khan et al., 2020). According to the Global Burden of Disease Study 2021, IHD accounted for 254.3 million prevalent cases (95% CI: 221.4, 295.5 million), approximately 9.0 million deaths (95% CI: 8.3, 9.5 million), and 188.4 million disability-adjusted life years (DALYs, 95% CI: 177.0, 198.2 million) worldwide in 2021 (Yang et al., 2025).

* Corresponding authors.
E-mail addresses: guanjun0439@163.com (G. Wang), zhangwj227@mail.sysu.edu.cn (W. Zhang), chenxinxin@gzhmu.edu.cn (X. Chen).
[1] These authors contributed equally to this work

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Besides well-known behavioral and biomedical risk factors (smoking, hypertension, etc.), environmental drivers such as fine particulate matter ($PM_{2.5}$) have been increasingly recognized as important contributors to IHD burden (Brook et al., 2017).

Accumulating epidemiological evidence has demonstrated a consistent association between $PM_{2.5}$ exposure and increased IHD mortality. For example, extensive cohort analyses and systematic reviews have uniformly indicated that sustained $PM_{2.5}$ exposure increases IHD mortality risk (hazard ratio (HR): 1.23 per 10 μg/m^3, 95% CI: 1.15, 1.31) (Alexeeff et al., 2021). Among cohorts of patients with pre-existing cardiovascular disease, the associations were more pronounced for IHD mortality than for other cardiovascular outcomes (HR: 1.82, 95% CI: 1.16, 2.83) (Xu et al., 2021). Data extracted from the American Cancer Society cohort further showed that intra-city $PM_{2.5}$ gradients were associated with IHD mortality (Relative Risk (RR): 1.24–1.6 per 10 μg/m^3) (Jerrett et al., 2005). Additionally, emerging findings extend these associations beyond aggregate $PM_{2.5}$ concentration, demonstrating varied toxicity of its chemical constituents, including elemental carbon (EC), organic carbon (OC), sulfate (SO_4^{2-}), nitrate (NO_3^-), and ammonium (NH_4^+) (Thurston et al., 2016). Specifically, per-interquartile range (IQR) increments in long-term exposure to EC, NH_4^+, NO_3^-, and SO_4^{2-} have been associated with significant IHD mortality elevations (HR ranging from 1.08 to 1.15) (Chen et al., 2025). Furthermore, SO_4^{2-} has been correlated with increased cardiovascular mortality (HR: 1.03, 95% CI: 1.00, 1.07) in a Danish cohort study (Hvidtfeldt et al., 2019). In addition, EC and OC have been linked to myocardial infarction mortality (Odds Ratio (OR): 1.038, 95% CI: 1.003, 1.074 for EC and OR: 1.057, 95% CI: 1.023, 1.097 for OC per IQR) (Mo et al., 2023). While the toxicity of $PM_{2.5}$ inherently stems from its complex mixture of components, current regulatory standards rely predominantly on total mass metrics. This mass-based approach is fundamentally limited because it can obscure the severe health impacts of highly toxic chemical constituents that comprise only a minor fraction of the total mass. Consequently, investigating component-specific mixtures is essential to precisely identify dominant toxic sources and inform targeted emission control policies (Esposito et al., 2016).

Despite the recognized importance of $PM_{2.5}$ composition, existing studies have several limitations. Most studies are geographically limited to certain regions or even cities, lacking nationwide coverage. Additionally, most studies focus on individual pollutants or limited groups of pollutants instead of $PM_{2.5}$ as a mixture (Lo et al., 2022; Hayes et al., 2020). From a methodological perspective, single-component models ignore the underlying interdependencies among $PM_{2.5}$ components. Due to collinearity issues, multi-pollutant studies cannot capture these intricate relationships and thus also fail to reflect synergistic effects. Although recent analyses have started to investigate mixture toxicity using source-apportioned or sector-specific approaches, (Berger et al., 2018) residual confounding and causal identification issues still exist. These issues impair the interpretation and policy relevance of the results from these studies. Hence, it is necessary to conduct a nationwide analysis that integrates advanced causal inference tools to quantify the joint effects of $PM_{2.5}$ component mixture on IHD mortality. First, compared to regional studies that often lack generalizability, a nationwide approach provides substantial statistical power and comprehensively captures the complex spatial variations of $PM_{2.5}$ components across diverse geographical areas. Second, some national studies use causal models for single pollutants, but assessing complex mixtures remains methodologically challenging. In contrast, leveraging a quasi-experimental design, such as the DID framework, utilizes spatial and temporal contrasts to effectively eliminate unmeasured time-invariant confounders, improving the reliability of effect estimates (Schwartz et al., 2021; Goodman et al., 2021; Li et al., 2023). Furthermore, by uniquely integrating this DID framework with a Quantile G-Computation (QGC) mixture model, our study not only evaluates the joint effect of combined exposure but also quantifies the relative hazard weights of specific components. This integrated framework aims to assess the potential health impacts of interventions, thereby providing reliable evidence for component-specific environmental policies (Zigler et al., 2016). Additionally, utilizing census-level data offers a critical methodological advantage over complex survey samples. Unlike survey-based designs where unweighted mixture analyses can yield biased associations and false positive findings due to disproportionate sampling, our total-population approach inherently avoids these validity threats.

To fill these gaps, this study examines IHD mortality associated with long-term annual exposure to $PM_{2.5}$ and its five main chemical components. Additionally, we employ Quantile G-Computation (QGC) approach within Difference-in-Differences (DID) framework to estimate the joint causal impact of $PM_{2.5}$ component mixture on the IHD mortality outcome of interest. We further explore possible effect modifications by seasonal temperature as well as demographic and socioeconomic characteristics to identify vulnerable subgroups.

2. Materials and methods

2.1. Outcome data

In this study, we used county-level mortality data for IHD covering 3108 counties across the contiguous United States from 2000 to 2019. These data were acquired from the United States Centers for Disease Control and Prevention (CDC) Wide-ranging Online Data for Epidemiologic Research (WONDER) database. The foundational death statistics originate from official death certificates recorded across all 50 U.S. states and Washington, D.C. These records are processed and standardized by the National Vital Statistics System (NVSS) under the National Center for Health Statistics (NCHS) before being made accessible via the WONDER online query system (Centers for Disease Control and Prevention, 2020). Mortality due to IHD was determined by International Classification of Diseases, Tenth Revision (ICD-10) codes I20-I25. All mortality data were further stratified by age group (<65 years and ≥65 years), sex (male and female), race (American Indian, Asian, Black or African American, and White), and Hispanic origin (Hispanic or Latino, and Not Hispanic or Latino) to enable subgroup analysis.

2.2. Exposure assessment

The analysis was based on population exposure to five major chemical components of $PM_{2.5}$, which includes EC, OC, SO_4^{2-}, NO_3^-, and NH_4^+ (Amini et al., 2022). These specific constituents were selected based on three criteria. First, they are currently available components with validated, high-resolution gridded data providing continuous nationwide coverage over the 20-year study period (2000–2019); Second, they collectively account for the majority (approximately 70–75%) of the total $PM_{2.5}$ mass (Philip et al., 2014); Third, their inclusion ensures methodological comparability with recent high-quality epidemiological studies evaluating $PM_{2.5}$ component mixtures (Chen et al., 2025; Jiang et al., 2024; Iyer et al., 2026). Gridded annual mean concentration data files for each component from 2000 to 2019 were obtained from the National Aeronautics and Space Administration Socioeconomic Data and Applications Center (SEDAC). Gridded annual mean concentration data files for each component from 2000 to 2019 were obtained from the National Aeronautics Space Administration and SEDAC. This dataset provided high-resolution estimates at 50 m resolution for urban areas and 1 km resolution for non-urban settings. To harmonize the spatial resolution, we aggregated the 50-m urban estimates to a 1-km grid and merged them with the non-urban estimates. Subsequently, we computed county-level annual mean concentrations by averaging 1-km grid cell values within each county boundary. Total $PM_{2.5}$ mass concentration data were obtained from a distinct SEDAC dataset and underwent the same spatial processing protocol (van Donkelaar et al., 2019). Through official email requests, we obtained the internally validated, updated concentration data extending through

2019 to ensure complete temporal coverage for our study period. To better represent exposure during the corresponding study year, we used the annual mean concentration of each component as the exposure variable in all models.

2.3. Covariate assessment

To account for potential confounding factors, we collected data on meteorological and socioeconomic variables. Hourly gridded temperature data for the contiguous U.S. (2000–2019) were sourced from the ERA5-Land dataset, which is provided by the European Centre for Medium-Range Weather Forecasts at ~9-km resolution. We calculated county-level daily mean temperatures by averaging the hourly grid values across space. These daily values were then used to derive seasonal averages and standard deviations for summer (June–August) and winter (December–February). Socioeconomic and health-related indicators included county-level prevalence of poverty, obesity, and physical inactivity, obtained from the Behavioral Risk Factor Surveillance System (BRFSS) (Centers for Disease Control and Prevention, 2024). Data on health insurance coverage and per capita personal income (PCPI) were sourced from the U.S. Bureau of Economic Analysis (BEA) (U.S. Bureau of Economic Analysis, 2024). Due to data availability, obesity and physical inactivity data were available for 2004–2019, and health insurance coverage data for 2006–2019.

2.4. Statistical analysis

Our study utilized a quasi-experimental difference-in-differences framework. We applied a poisson regression incorporating two-way fixed effects to evaluate the extended causal relationship between yearly $PM_{2.5}$ constituent levels and IHD death rates. This framework effectively controls for time-invariant county-specific characteristics through county fixed effects and adjusts for secular trends common to all counties via year fixed effects. This design isolates the effect of spatio-temporal variations in air pollutant exposure, thereby mitigating potential bias from unobserved confounders. To improve model performance and ensure consistency with previous studies, all models were adjusted for key time-varying covariates, specifically PCPI and seasonal temperatures (Wang et al., 2016, 2023; Renzi et al., 2019). The refined two-way fixed effects(TWFE) model is therefore specified as:

$$\begin{aligned}\ln[E(Y_{s,t})] &= \beta_0 + \beta_1 I_s + \beta_2 I_t + \beta_3 Exposure^i_{s,t} + \beta_4 Smean_temp_{s,t} \\ &+ \beta_5 Wmean_temp_{s,t} + \beta_6 Ssd_temp_{s,t} + \beta_7 Wsd_temp_{s,t} + \beta_8 PCPI_{s,t} \\ &+ \beta_9 Reidual_{s,t} + \ln(Pop_{s,t}),\end{aligned}$$

where I_s and I_t were indicator variables for county s and year t, respectively. $Smean_temp_{s,t}$, $Wmean_temp_{s,t}$, $Ssd_temp_{s,t}$, and $Wsd_temp_{s,t}$ were the mean and standard deviation of summer and winter temperatures, respectively. $PCPI_{s,t}$ represented the PCPI per capita in county s and year t, and ln $(Pop_{s,t})$ was an offset term.

The associations were reported as the percent increase in risk of IHD mortality (IR%) per IQR increment in each pollutant's concentration. The IR% was calculated as: IR% = (RR - 1) × 100%, where RR represents the relative risk estimated from the model. To account for confounding from the remaining particulate matter, each single-pollutant model was adjusted for the residual $PM_{2.5}$ mass (i.e., total $PM_{2.5}$ concentration minus the concentration of the specific constituent being assessed).

Further, to evaluate the cumulative impact of mixed exposure to multiple components and to address multicollinearity among the $PM_{2.5}$ components, we integrated QGC into the DID framework. This method allows for the simultaneous estimation of mixture effects and component weights, specified as:

$$Y = \beta_0 + \psi \sum_{i=1}^{d} w_i X_i^q + \varepsilon = \beta_0 + \sum_{i=1}^{d} \beta_i X_i^q + \varepsilon,$$

where X_i^q represented quantized version of the ith component. w_i was the weight of component i; ψ was the overall effect of component mixture in WQS model. The overall effect of component mixture in QGC was $\psi = \sum_{i=1}^{d} \beta_i$.The QGC framework provides several methodological advantages: It could eliminate the requirement for unidirectional effect assumptions, accommodates non-linear relationships, and enables the incorporation of interaction terms between mixture components or between components and covariates (Keil et al., 2020). Following primary model specification, counterfactual datasets were generated through model-based prediction. Population-level mixture effects were subsequently estimated using marginal structural models (MSMs). Finally, the QGC model combined with DID design in this study was constructed as follows:

(1) Primary model

$$\begin{aligned}\ln[E(Y_{s,t})] &= \beta_0 + \beta_1 I_s + \beta_2 I_t + \sum_{i=1}^{5} \gamma_i Component^q_{s,t,i} \\ &+ \beta_3 Smean_{temp_{s,t}} + \beta_4 Wmean_temp_{s,t} + \beta_5 Ssd_temp_{s,t} \\ &+ \beta_6 Wsd_temp_{s,t} + \beta_7 PCPI_{s,t} + \ln(Pop_{s,t}),\end{aligned}$$

(2) MSM

$$\ln[E(Y_{s,t})] = \beta_0 + \psi Mixture_{s,t}.$$

The QGC method could assess the joint effect of the $PM_{2.5}$ component mixture and quantify the weights of each component (Keil et al., 2020). Specifically, components with positive regression coefficients are normalized to yield positive weights that sum to 1.0, representing their relative contribution to the positive mixture effect; similarly, components with negative coefficients are separately normalized to yield negative weights that also sum to 1.0, indicating their relative contribution to the negative mixture effect. The overall net effect of the exposure mixture represents the sum of these positive and negative directional effects. Therefore, a positive weight indicates a relative contribution to the overall positive mixture effect (e.g., increasing IHD mortality) rather than an absolute effect size, and should be strictly interpreted as an indicator of relative importance rather than an independent pathogenic determination. We also investigated potential effect modification by various factors by introducing interaction terms into the models. The modifiers examined included demographic (e.g., age, sex, race, and ethnicity), socioeconomic (e.g., poverty, physical inactivity, and insurance coverage), and climatic (e.g., seasonal temperature) factors. The statistical significance of any differences in effect estimates between subgroups was formally evaluated using meta-regression.

2.5. Sensitivity analysis

A series of sensitivity analyses were conducted to confirm the robustness of our primary findings. First, based on the main model, we further adjusted for the proportion of population aged ≥ 65 years, females, White, Hispanic or Latino, and poverty rate as Additional Model 1, for which data were complete from 2000 to 2019. We then constructed Additional Model 2, which additionally included physical inactivity rate, obesity rate, and health insurance coverage rate. Second, we repeated the mixture analysis using quintile-and decile-based pollutant exposure distributions instead of quartiles to test model stability. Third, E-values were computed to assess the minimum strength of association that an unmeasured confounder would require with both exposure and outcome to fully explain away observed associations. Finally, because the DID framework relies on the parallel trends assumption, we indirectly evaluated its plausibility by comparing

temporal changes in $PM_{2.5}$ components, PCPI, and temperature across the study period, consistent with previous literature. All statistical analyses were conducted in R version 4.2.2.

3. Results

3.1. Descriptive analysis

From 2000–2019, 8162,305 U.S. deaths were attributed to IHD, corresponding to an average annual mortality rate of 133.9 per 100,000. IHD mortality varied markedly across demographic subgroups. The highest mortality occurred among older adults (aged ≥65 years; 795.65 per 100,000 population), males (147.55), White individuals (146.22), and non-Hispanic or Latino individuals (150.54) (Table 1). Descriptive statistics for $PM_{2.5}$ components and covariates are provided in Supplementary Table 1. IHD mortality declined steadily and substantially from 2000 to 2019 (Fig. 1**B**). The decline was evident across all subgroups by age, sex, race, and ethnicity (Fig. 1**C-F**). Concurrently, the annual mean concentrations of all 5 $PM_{2.5}$ components showed a marked decline over the same period (Supplementary Figure 1). Strong positive correlations occurred between EC and OC, EC and NO_3^-, and NO_3^- and NH_4^+ (Supplementary Table 2). Geographically, the highest county-level IHD mortality was clustered in the Southern and Appalachian regions (Fig. 1**A**). In contrast, higher concentrations of $PM_{2.5}$ components were mainly found in the Midwest and South (Supplementary Figure 2).

3.2. Association of $PM_{2.5}$ component exposure with IHD mortality

Single-component models showed that each $PM_{2.5}$ component was significantly associated with higher IHD mortality, with IQR increases linked to 1.61–8.83% higher risk (Table 2). We assessed the main effect of total $PM_{2.5}$ mass concentration. Our model estimated that 1 IQR increase in total $PM_{2.5}$ mass was significantly associated with a 1.41% increase in IHD mortality risk (95% CI: 0.93, 1.90). In the joint effect model, each IQR increase in the $PM_{2.5}$ component mixture was associated with a 0.92% (95% CI: 0.60, 1.25) increase in IHD mortality risk. Within the $PM_{2.5}$ component mixture, SO_4^{2-} (weight: 43.69%) and NO_3^- (weight: 34.97%) contributed the largest positive weights to the joint effect, whereas EC exhibited a negative weight (Fig. 2). Additionally, nonlinear exposure–response relationships were observed between individual $PM_{2.5}$ components and IHD mortality, whereas no evidence of nonlinearity was detected for the $PM_{2.5}$ component mixture. (Supplementary Figure 3).

Table 1
Descriptive statistics of ischemic heart disease deaths and mortality rates by demographic characteristics in the United States during 2000–2019.

Variable	Deaths from ischemic heart disease, N (%)	Ischemic heart disease mortality rate (per 100,000 population)
All participants	8162,305 (100.00)	133.87
Age		
< 65 years	1535,515 (18.81)	29.17
≥ 65 years	6626,252 (81.18)	795.65
Not Stated	538 (0.01)	-
Sex		
Male	4422,891 (54.19)	147.55
Female	3739,414 (45.81)	120.65
Race		
White	7113,948 (87.16)	146.22
American Indian	37,169 (0.46)	47.57
Asian	145,017 (1.78)	45.41
Black or African American	866,171 (10.61)	103.82
Hispanic Origin		
Not Hispanic or Latino	7704,925 (94.40)	150.54
Hispanic or Latino	431,230 (5.28)	44.06
Not Stated	26,150 (0.32)	-

Abbreviations: N: number.

3.3. Effect modification by seasonal temperature

Seasonal temperature modified the association between the $PM_{2.5}$ component mixture and IHD mortality (Table 3). Compared to the median (50th percentile) temperatures, the mortality risks were significantly greater at the 75th percentile for both summer ($P_{interaction} = 0.04$) and winter ($P_{interaction} < 0.01$). In summer, the mortality risk was 0.51% (95% CI: 0.15, 0.86) at the 25th percentile (P25, $P_{interaction} = 0.08$), increased to 0.94% (95% CI: 0.61, 1.27) at the median (P50), and rose to 1.45% (95% CI: 1.10, 1.80) at the 75th percentile (P75). A similar increasing trend was observed in winter, where the risk escalated from an inverse association of −1.35% (95% CI: −1.73, −0.98) at P25 ($P_{interaction} < 0.01$), to 0.09% (95% CI: −0.25, 0.42) at P50, and reached 1.56% (95% CI: 1.22, 1.89) at P75. These results indicate that elevated temperatures across seasons consistently intensified the cardiovascular impacts of the $PM_{2.5}$ mixture.

3.4. Analysis of potential modifying effects

The strength of the association between the $PM_{2.5}$ component mixture and IHD mortality varied greatly among population subgroups (Fig. 3). Compared with their respective reference groups, the mortality risk associated with the $PM_{2.5}$ component mixture increased by 4.81% (95% CI: 3.92, 5.70) for older adults aged ≥ 65 years and by 3.83% (95% CI: 2.24, 5.45) for Hispanic or Latino populations. In contrast, no significant association was found among females (IR%: −0.32; 95% CI: −0.96, 0.33), and the effect estimate for Black or African American populations was 2.46% lower than White populations (95% CI: −3.84, −1.05). With regard to socioeconomic factors, compared to their respective reference counties, the adverse effect of the $PM_{2.5}$ component mixture was significantly increased by 0.30% (95% CI: 0.15, 0.44) among residents of counties with higher rates of poverty and by 0.92% (95% CI: 0.79, 1.05) in counties with higher physical inactivity rates. Additionally, areas with higher obesity prevalence showed a non-significant positive association (IR%: 0.13; 95% CI: 0.00, 0.27). Conversely, counties with higher health insurance coverage showed a significant 1.05% risk reduction (95% CI: −1.23, −0.87) compared to counties with lower coverage.

3.5. Sensitivity analysis

We performed multiple sensitivity analyses to assess robustness. First, models with additional adjustment for other covariates yielded results consistent with the main model; notably, because data for certain covariates were limited to later years, these models confirmed that the main effects held steady across the restricted 2004–2019 and 2006–2019 sub-cohorts (Supplementary Table 3, Supplementary Table 4). Furthermore, temporal stratification revealed a stronger joint mixture effect in the later decade, with the IR% increasing from 0.19% (95% CI: −0.37, 0.75) during 2000–2009–1.43% (95% CI: 0.89, 1.98) during 2010–2019 (Supplementary Table 5). Second, pollutant concentrations were categorized into quintiles and deciles, results were consistent with the main analysis, and the resulting trends were consistent with those from the primary analysis (Supplementary Table 6). Furthermore, correlations between temporal changes in pollutants and key covariates were low (all correlation coefficients < 0.31), suggesting that the quasi-experimental design assumption was met (Supplementary Table 7). We primarily adjusted for summer and winter temperatures to capture extreme weather stress while avoiding multicollinearity, as spring and autumn temperatures are highly correlated with other seasons, whereas summer and winter exhibit the lowest correlation (Supplementary Table 8). To further address potential meteorological confounding during transitional seasons, we conducted a

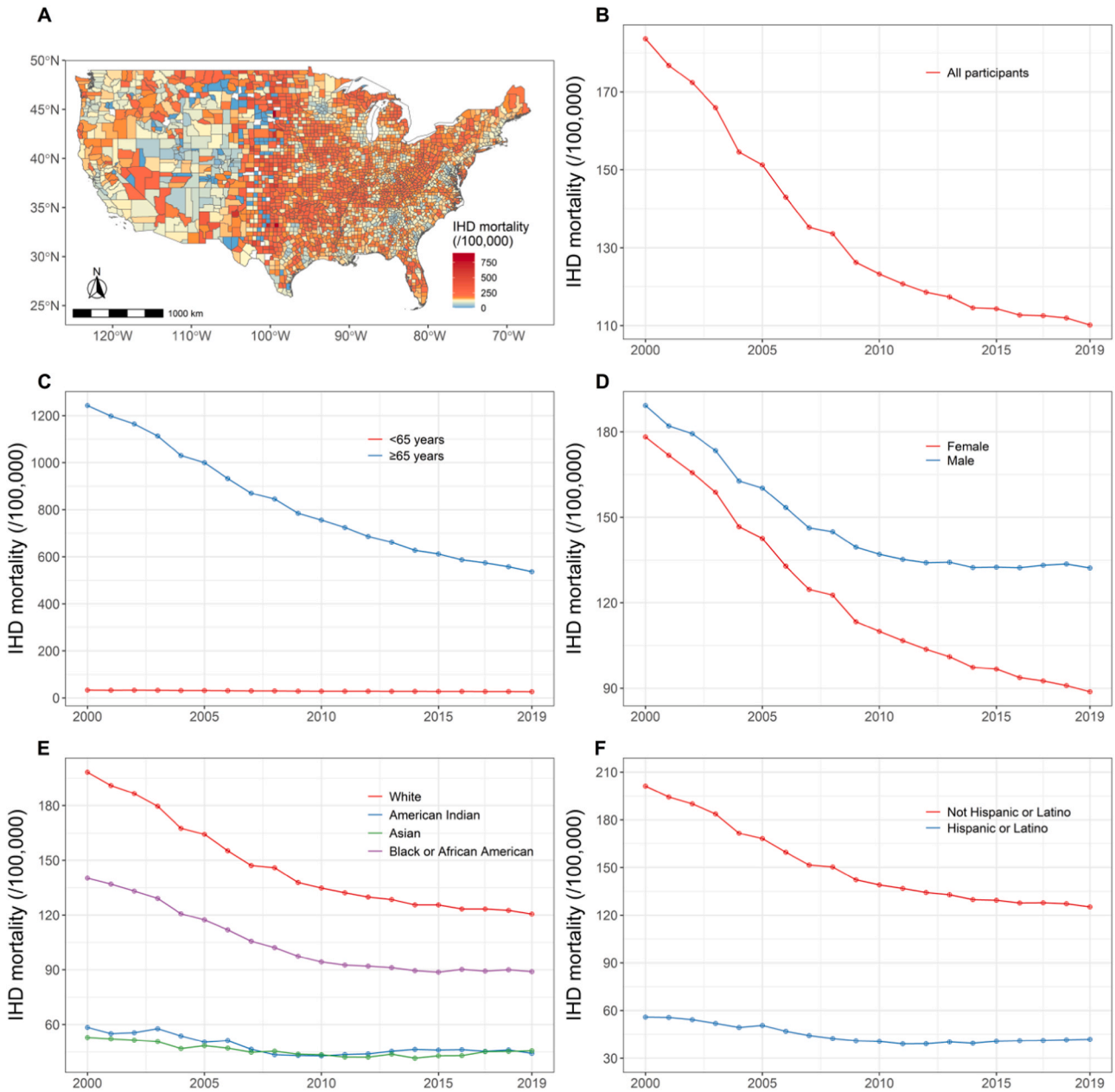

Fig. 1. Spatial (A) and temporal (B-F) trends in ischemic heart disease mortality among all participants (B), stratified by age (C), sex (D), race (E), and Hispanic origin (F) in the United States from 2000 to 2019. Abbreviations: IHD: ischemic heart disease.

Table 2
Causal effects of an interquartile range (IQR) increase in $PM_{2.5}$ components and their component mixture on ischemic heart disease mortality.

Component	Ischemic heart disease mortality	
	IR% (95% CI)	*P*
Total $PM_{2.5}$ Mass	1.41 (0.93, 1.90)	< 0.01
Elemental carbon	8.83 (8.21, 9.45)	< 0.01
Organic carbon	1.61 (1.03, 2.20)	0.014
Sulfate	1.72 (1.21, 2.24)	< 0.01
Nitrate	3.72 (3.20, 4.25)	< 0.01
Ammonium	3.31 (2.68, 3.93)	< 0.01
Component mixture	0.92 (0.60, 1.25)	< 0.01

Abbreviations: IR%: percent increase in risk of IHD mortality; CI: confidence interval; $PM_{2.5}$: fine particulate matter; *P*: *P*-value.

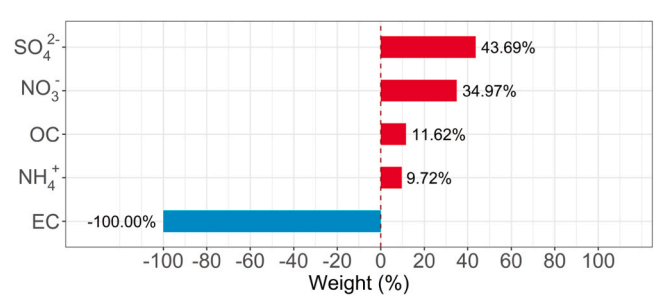

Fig. 2. Weights of each component in the association between the $PM_{2.5}$ component mixture and ischemic heart disease mortality. Abbreviations: $PM_{2.5}$: fine particulate matter; EC: elemental carbon; OC: organic carbon; SO_4^{2-}: sulfate; NO_3^-: nitrate; NH_4^+: ammonium.

Table 3
Effect modification of the association between the $PM_{2.5}$ component mixture and ischemic heart disease mortality by seasonal temperature.

Temperature	Summer temperatures		Winter temperatures	
	IR% (95% CI)	P for interaction	IR% (95% CI)	P for interaction
P50	0.94 (0.61, 1.27)		0.09 (-0.25, 0.42)	
P25	0.51 (0.15, 0.86)	0.08	-1.35 (-1.73, -0.98)	< 0.01
P75	1.45 (1.10, 1.80)	0.04	1.56 (1.22, 1.89)	< 0.01

* Indicates a statistically significant difference compared to the effect estimate at the 50th percentile (P50) temperature.
Abbreviations: $PM_{2.5}$: fine particulate matter; P25: 25th percentile; P50: 50th percentile (median); P75: 75th percentile; CI: confidence interval.

sensitivity analysis additionally incorporating the means and standard deviations of spring and autumn temperatures into the TWFE model. The results confirmed that the main causal links between the $PM_{2.5}$ component mixture and IHD mortality remained highly robust after this comprehensive seasonal adjustment (Supplementary Table 9). We also evaluated the models using full-year annual mean temperatures instead of seasonal temperatures. The main effects remained robust (Supplementary Table 10), and the effect modification analysis confirmed that higher annual temperatures significantly exacerbated the mortality risk associated with the mixture (Supplementary Table 11). Finally, E-values ranged from 1.11 to 1.40, suggesting limited potential impact of unmeasured confounding (Supplementary Table 12).

4. Discussion

This study utilized nationwide, county-level data across the contiguous United States from 2000 to 2019, applying a causal inference framework that integrated difference-in-differences and quantile g-computation method to assess the associations between annual exposure to $PM_{2.5}$ components and IHD mortality. We found that both individual $PM_{2.5}$ components and their mixtures were associated with increased risks of IHD mortality, with SO_4^{2-} and NO_3^- contributing the largest positive weights to the mixture effect. In the joint effects of $PM_{2.5}$ component mixture, higher risks were observed among older adults, Hispanic or Latino individuals and White populations, and in counties with higher poverty and physical inactivity rates and lower health insurance coverage.

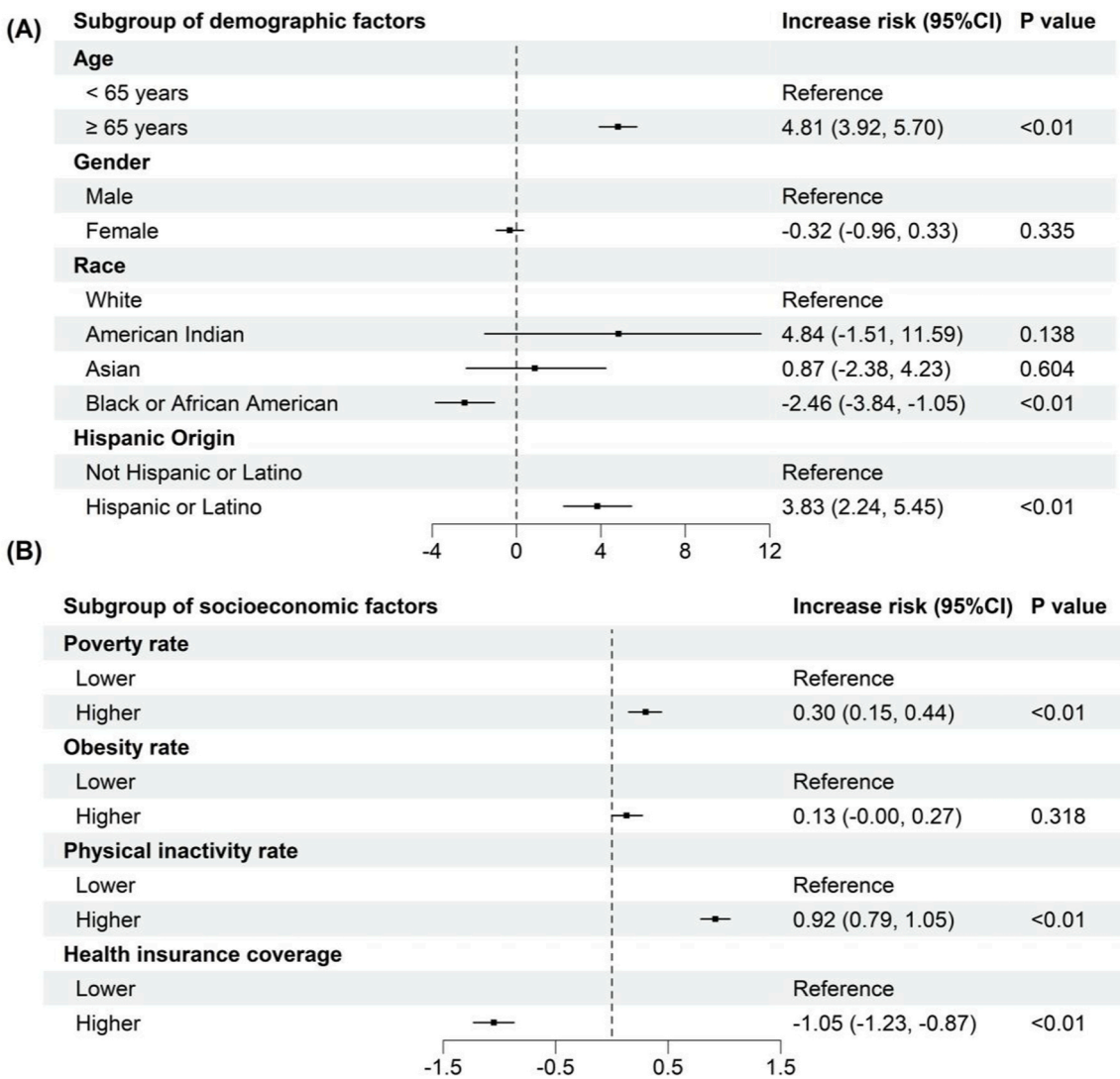

Fig. 3. Results on effect modification of the association between $PM_{2.5}$ component mixture and ischemic heart disease mortality demographic by (A) characteristics and (B) socioeconomic factors. [a] Percent risk increase across variable subgroups relative to reference groups. Abbreviations: $PM_{2.5}$: fine particulate matter; CI: confidence interval; *P*: *P*-value.

At the national level, we found that specific $PM_{2.5}$ chemical constituents and their joint mixture are associated with increased IHD mortality, which is aligned with multiple high-quality studies. Nationwide source-apportionment analyses have linked source-related constituents to elevated IHD mortality (Thurston et al., 2016). Large U.S. cohort analyses have similarly reported constituent-specific associations with cause-specific mortality, strengthening the evidence that toxicity varies across $PM_{2.5}$ components (Kazemiparkouhi et al., 2022). More recent work from diverse settings continues to identify particular chemical species as important drivers of cardiovascular risk, underscoring the relevance of examining chemical composition (Zhou et al., 2024; Tian et al., 2024). Importantly, whereas these prior studies establish constituent-level hazards, fewer have estimated joint mixture effects using a formal mixture framework; by applying QGC within a causal DID design, our study quantifies the combined contribution of components and thus complements the existing literature (Zheng et al., 2025). The biological pathways linking $PM_{2.5}$ components to IHD are considered multifactorial, with systemic inflammation and oxidative stress constituting core mechanisms (Brook et al., 2017). Inhaled fine particles can translocate into the bloodstream, triggering the release of pro-inflammatory cytokines and promoting endothelial dysfunction, a key initiating step in atherosclerosis. This results in oxidative stress, damaging lipids, proteins, and DNA, thereby contributing to plaque instability and thrombosis, which may culminate in acute coronary events such as myocardial infarction (Zanobetti et al., 2010).

Furthermore, we observed that elevated seasonal temperatures, both during hotter summers and anomalously warm winters, significantly exacerbated the mortality risk associated with the $PM_{2.5}$ component mixture. This temperature-driven amplification of toxicity can be elucidated by three primary mechanisms. First, extreme meteorological conditions have been shown to exert a bi-directional modifying effect on the cardiovascular impacts of $PM_{2.5}$ (Hu et al., 2025). Second, ambient temperature fluctuations perturb atmospheric physicochemical reactions, altering the generation and composition of highly toxic secondary aerosols within the mixture, which directly elevates overall toxicity (Elamouri et al., 2018). Third, extreme temperatures increase the human thermoregulatory burden, inducing systemic physiological stress that renders the cardiovascular system inherently more susceptible to $PM_{2.5}$-induced oxidative stress and lipid metabolism abnormalities (Elamouri et al., 2018; Coca et al., 2023). Conversely, the inverse associations observed at extreme low temperatures (P25) do not imply biological protection. Instead, they likely result from severe exposure misclassification, as populations spend more time indoors during extreme cold, significantly reducing true personal exposure to outdoor $PM_{2.5}$, and potential competing mortality risks from severe cold stress itself (Alexeeff et al., 2023; Kioumourtzoglou et al., 2016; Castro et al., 2025).

Notably, in the $PM_{2.5}$ component mixture, SO_4^{2-} and NO_3^- exhibited the largest positive weights, indicating that secondary inorganic aerosols may be key drivers of the observed association with IHD mortality. This finding is consistent with previous epidemiological studies reporting stronger cardiovascular mortality risks associated with sulfate-and nitrate-rich $PM_{2.5}$ compared with total mass alone (Weichenthal et al., 2024; Zhang et al., 2025). These components are primarily formed through atmospheric oxidation of SO_2 and NO_X from fossil fuel combustion and traffic emissions and have been linked to heightened oxidative stress and systemic inflammation (Zhao et al., 2018). Sulfate can enhance particle acidity and promote metal-catalyzed reactive oxygen species generation, while nitrate exposure has been associated with pro-inflammatory responses and endothelial dysfunction-pathways central to atherosclerosis and ischemic heart disease development. Together, existing evidence supports our observation that sulfate and nitrate disproportionately contribute to the cardiovascular toxicity of $PM_{2.5}$ mixtures, highlighting the importance of targeting precursor emissions in air quality control strategies.

Interestingly, our temporal stratification revealed a stronger joint effect of the $PM_{2.5}$ component mixture on IHD mortality during 2010–2019 compared to 2000–2009, despite a nationwide decline in overall $PM_{2.5}$ mass and dominant constituents like sulfate. This seemingly paradoxical phenomenon may be driven by two interconnected mechanisms. First, robust epidemiological evidence suggests a supralinear concentration-response relationship for $PM_{2.5}$ exposure, where the marginal increase in cardiovascular mortality risk is paradoxically steeper at lower concentration ranges (Weichenthal et al., 2022; Burnett et al., 2018). Consequently, as overall air quality improves, the per-unit toxicity of the remaining $PM_{2.5}$ mixture may appear amplified. Second, strict emission mitigation strategies over the past two decades have fundamentally altered the physicochemical composition of the $PM_{2.5}$ mixture. While coal-combustion-related sulfate decreased rapidly, our data indicates that the overall cardiovascular toxicity profile is increasingly driven by traffic- and combustion-related EC, as well as agriculture-associated ammonium and nitrate. These specific constituents possess high oxidative potential and have been shown to exert disproportionate cardiovascular toxicity (Jiang et al., 2024), highlighting that future clean air policies must shift focus from total mass reduction to targeting specific high-toxicity emission sources.

In addition, we observed a negative weight for EC in the joint mixture effect. A previous study conducted in the United States reported a similar finding (Wu et al., 2025). This negative weight should not be interpreted as evidence of an inverse association between EC and IHD mortality; rather, it likely reflects model convergence behavior in the presence of highly correlated mixture components (Keil et al., 2020). Furthermore, EC was strongly correlated with OM (r: 0.780) and NO_3^- (r: 0.744). Consequently, the positive association between EC and IHD mortality observed in single-pollutant models may be confounded by co-exposure to OM and NO_3^-. Further studies using alternative mixture modeling approaches and independent data sources are warranted to clarify the independent role of EC in IHD mortality.

Our findings suggest that the association between the $PM_{2.5}$ component mixture and IHD mortality exhibits health disparities, with certain susceptible subpopulations experiencing higher risks. The much greater risk among older adults could be mediated by immunosenescence, weakened antioxidant defense, and cumulative atherosclerotic burden, which render older adults more susceptible to systemic inflammation and vascular dysfunction after exposure to pollution (Chanda et al., 2024). The greater sensitivity among Hispanic or Latino adults may be due to their greater burden of obesity and diabetes, as well as potentially higher exposure from closer residential proximity to traffic and industrial sources. Furthermore, chronic low-grade inflammation, insulin resistance, and endothelial dysfunction caused by the aforementioned metabolic disorders likely increase their IHD risk (Rajagopalan et al., 2024). By contrast, we found that the excess risk attributable to the $PM_{2.5}$ component mixture was substantially reduced among Black or African American populations, which is consistent with several previous reports. This may in part be due to more advantageous lipid profiles among Black or African American study participants (i.e., higher high-density lipoprotein cholesterol and lower triglycerides), which may reduce the $PM_{2.5}$-induced pathways of inflammation and oxidative stress leading to IHD (van den Brekel et al., 2024; Sumner, 2009). Women in this study showed an inverse association that was not significant; however, previous studies have suggested that the sex-specific hormonal regulation of inflammation and vascular function, as well as different occupational and lifestyle exposures, may place women at a greater or different risk such that the sex effect is more complex than expected and more nuanced stratified analyses are warranted (Krittanawong et al., 2023). Communities of lower socioeconomic status face significantly higher risks, potentially due to poor nutrition, inadequate housing, chronic psychosocial stress, and limited access to medical resources—all of which may exacerbate pollution-related harms (Chi et al., 2016). Residents of high-inactivity communities may face greater cardiovascular risk from ambient $PM_{2.5}$ exposure because their habitual inactivity limits endothelial repair and

anti-inflammatory responses, thereby exacerbating particulate-induced vascular injury (Hao et al., 2025). Although obesity did not significantly modify effects in this study, a large body of studies demonstrated that obesity would increase $PM_{2.5}$ toxicity by promoting chronic low-grade inflammation, insulin resistance, and oxidative stress, and thus may become an important susceptibility factor (Weichenthal et al., 2014). Finally, higher health insurance coverage attenuated the association between the $PM_{2.5}$ component mixture and IHD mortality. This may be due to better chronic disease control and timely access to acute cardiac care, emphasizing the important role of public health systems in mitigating environmental risks (Wang et al., 2024).

First, the use of nationwide data from 2000 to 2019 provided complete population coverage rather than a representative sample. The validity and robustness of this nationwide county-level study design, utilizing comprehensive CDC WONDER mortality records and high-resolution air pollution data, have been well-established in previous high-impact investigations (Feng et al., 2024; Danesh Yazdi et al., 2024; Yin et al., 2025). Second, we employed a novel QGC model in DID causal inference framework. The model not only strictly controlled for time-invariant unmeasured confounders, but also successfully solved high collinearity among $PM_{2.5}$ components and obtained more stable and reliable mixture effect estimates. Third, our detailed effect modification analyses provided a comprehensive risk profile, which identified high-risk populations and climatic conditions.

Several limitations warrant consideration. First, this is an ecological study using county-level aggregated data. While ecological fallacy is a potential concern, our quasi-experimental DID framework mitigates this by utilizing spatial and temporal contrasts to eliminate both time-varying and time-invariant confounders, thereby providing the effect estimates with causal implications. Second, county-level exposure aggregation inevitably introduces non-differential measurement error, which typically biases estimates toward the null; thus, our findings likely represent conservative estimates of true individual-level risks (Bateson et al., 2007; Rich et al., 2019; Hopke et al., 2019). Third, our study did not evaluate trace metals. We restricted our focus to five core components that constitute the vast majority of total $PM_{2.5}$ mass, ensuring reliable long-term data coverage and methodological comparability with recent epidemiological literature. Fourth, the high-resolution gridded datasets for these specific $PM_{2.5}$ components are currently only available as annual averages, inherently lacking monthly or seasonal measurements. This objective data limitation precludes analyzing season-specific effects. To help alleviate this concern, we adjusted for seasonal temperatures in an effort to mitigate potential meteorological confounding.

5. Conclusion

In conclusion, this comprehensive nationwide study suggests potential causal links of exposure to $PM_{2.5}$ components and their component mixture with increased IHD mortality in the U.S., with SO_4^{2-} and NO_3^- emerging as the most toxic constituents. These risks are greater under higher temperatures and disproportionately borne by vulnerable subgroups, including older adults, Hispanic or Latino populations, residents of higher poverty rates and lower health insurance coverage, while this risk was decreased among Black or African American populations. Our findings underscore the limitations of regulations based solely on $PM_{2.5}$ total mass and highlight the urgent need for policies targeting specific emission sources enriched in toxic components, as well as public health strategies aimed at protecting populations most susceptible to the cardiovascular harms of air pollution.

CRediT authorship contribution statement

Zhiqiang Li: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xiaojing Song:** Data curation, Conceptualization. **Wangjian Zhang:** Writing – review & editing, Validation, Supervision, Software. **Andong Huang:** Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xinxin Chen:** Writing – review & editing, Validation, Funding acquisition, Formal analysis, Data curation. **Yanji Qu:** Supervision, Methodology, Investigation. **Lin Shao:** Methodology, Data curation. **Jiaqi Wei:** Methodology, Data curation. **Man Zhang:** Supervision, Formal analysis, Conceptualization. **Guanjun Wang:** Writing – review & editing, Visualization, Data curation, Conceptualization. **Qiao Hu:** Visualization, Validation, Supervision. **Guanghui Dong:** Supervision, Software, Project administration.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2026.120108.

Data availability

Data will be made available on request.

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